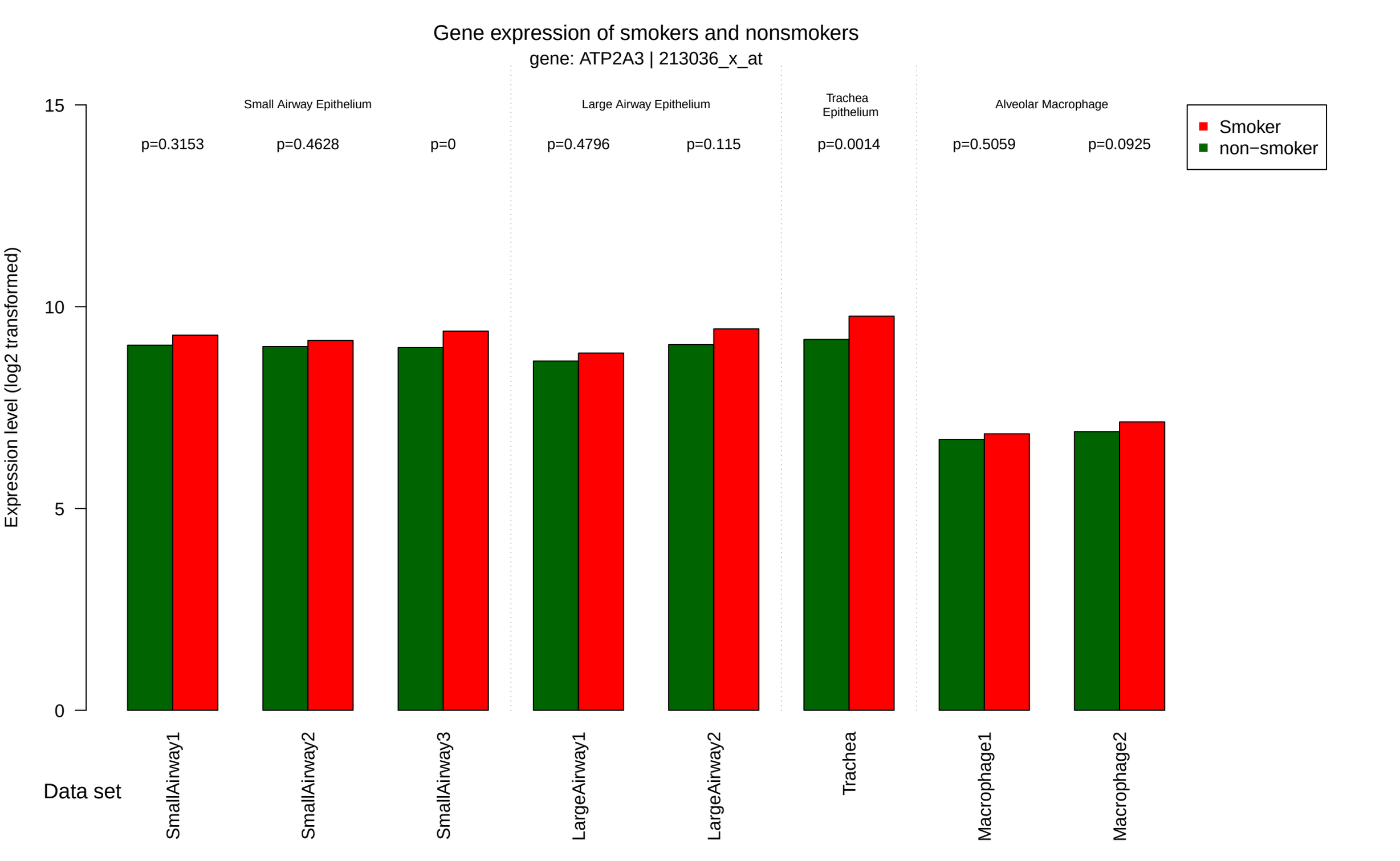
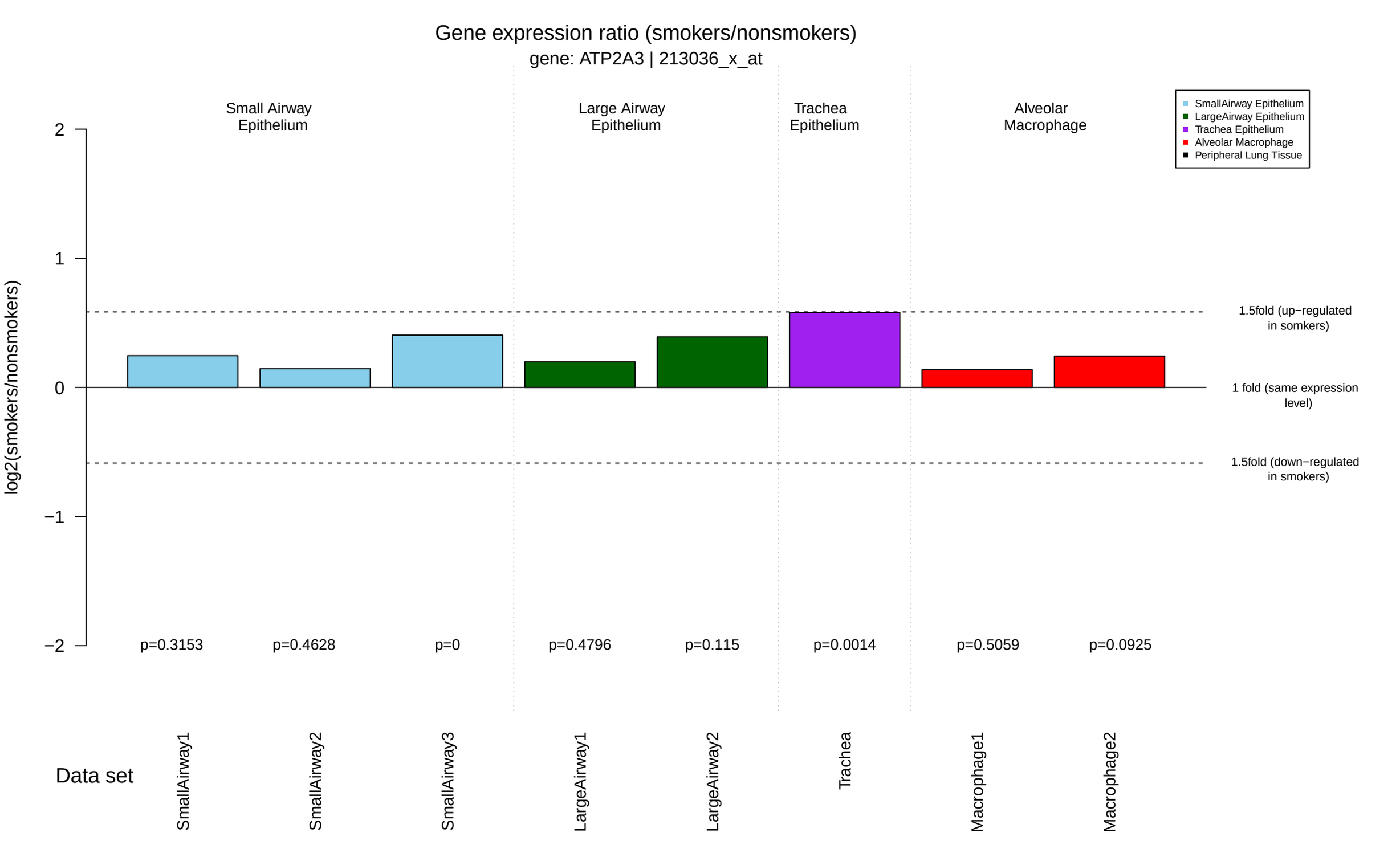
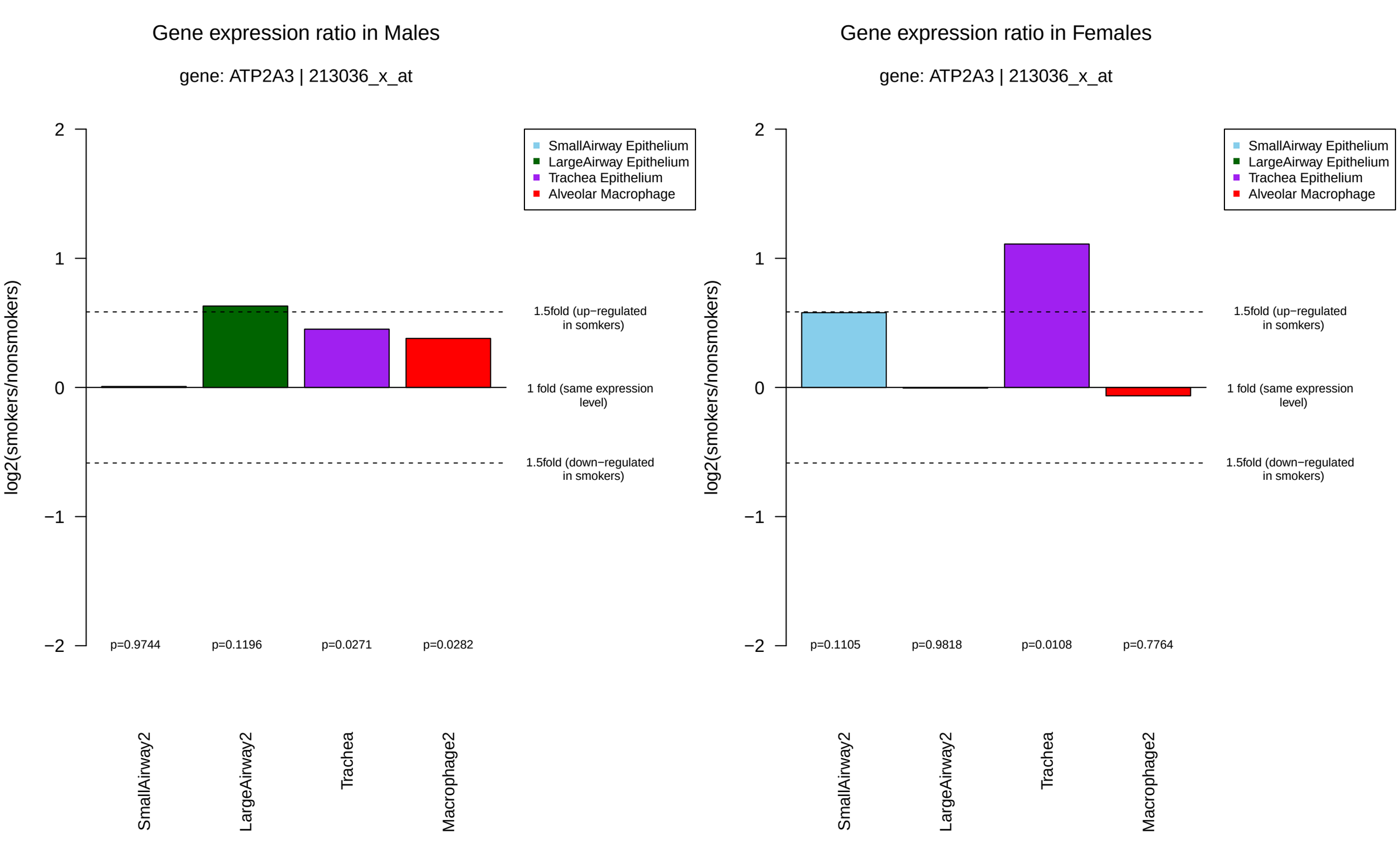
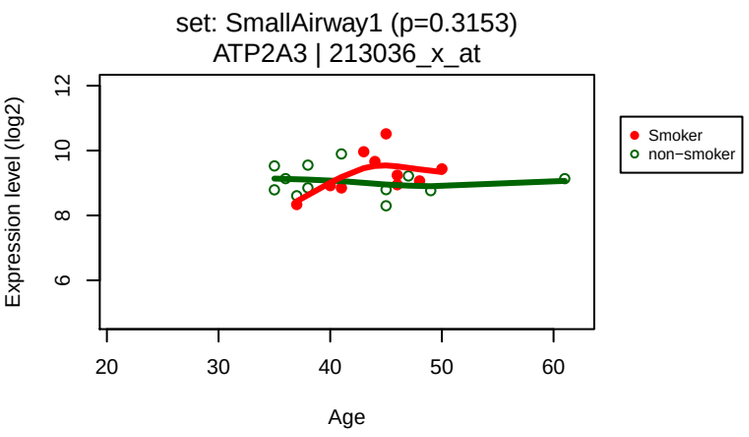
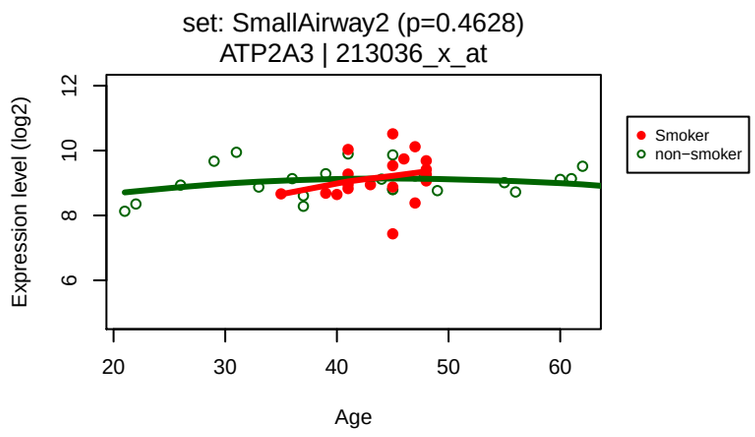
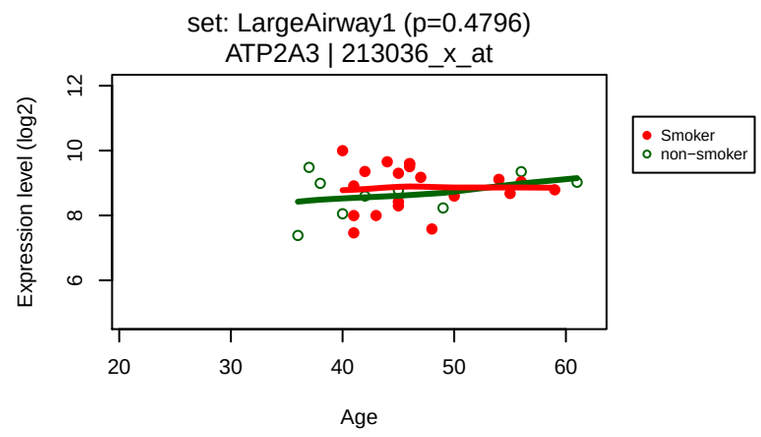

Gene expression of smokers and nonsmokers
gene: ATP2A3 | 213036_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3153
p=0.4628
p=0
p=0.4796
p=0.115
p=0.0014
p=0.5059
p=0.0925
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

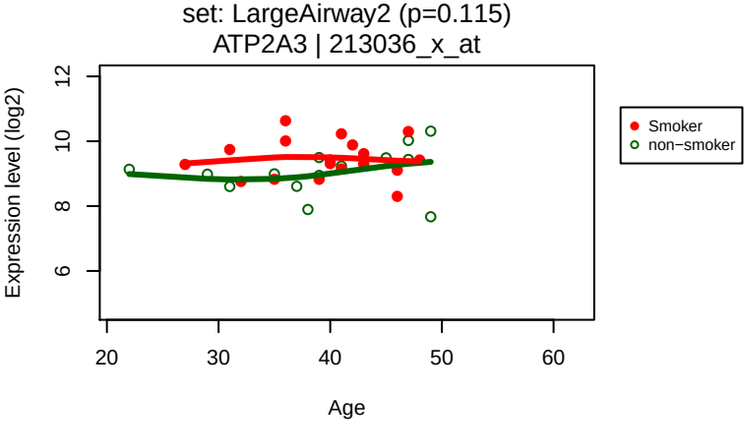

Gene expression ratio (smokers/nonsmokers)
gene: ATP2A3 | 213036_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3153
p=0.4628
p=0
p=0.4796
p=0.115
p=0.0014
p=0.5059
p=0.0925
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

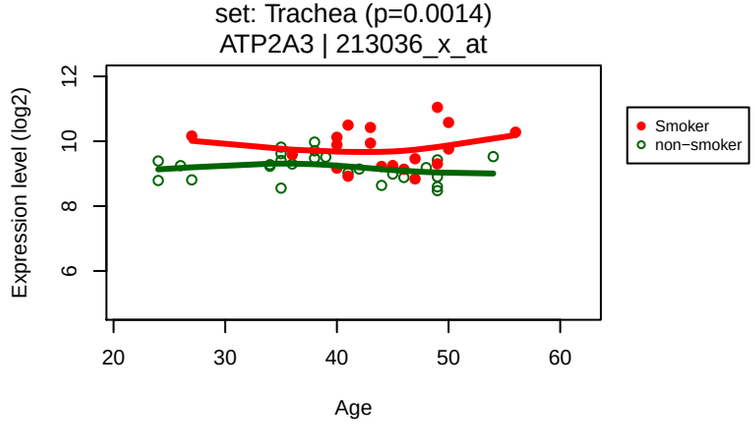

Gene expression ratio in Males
gene: ATP2A3 | 213036_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9744
p=0.1196
p=0.0271
p=0.0282
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP2A3 | 213036_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1105
p=0.9818
p=0.0108
p=0.7764
SmallAirway2
LargeAirway2
Trachea
Macrophage2

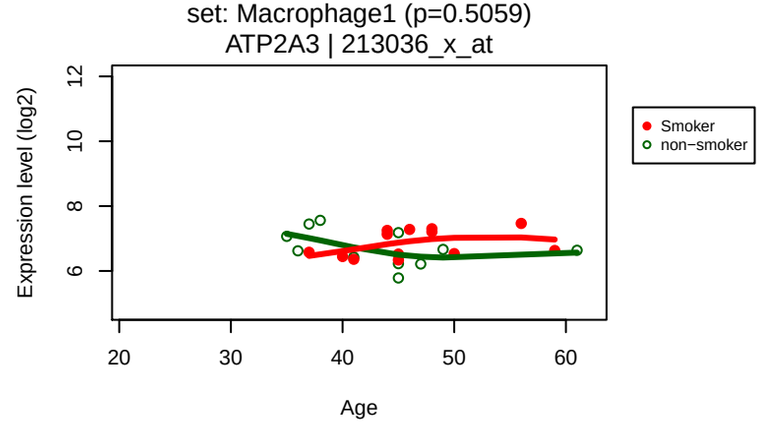

set: SmallAirway1 (p=0.3153)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker

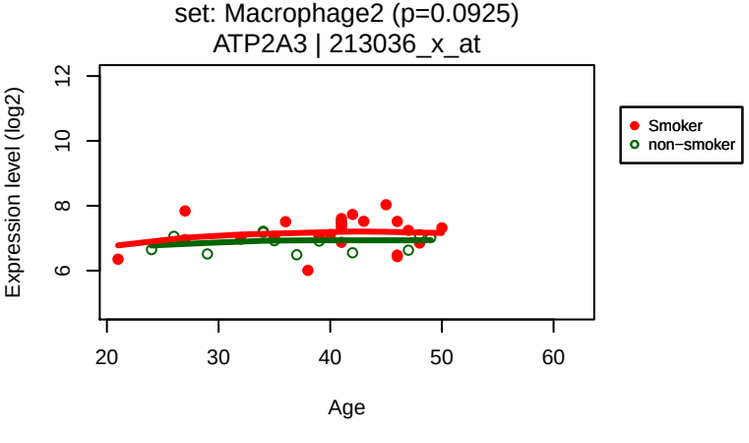

set: SmallAirway2 (p=0.4628)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker

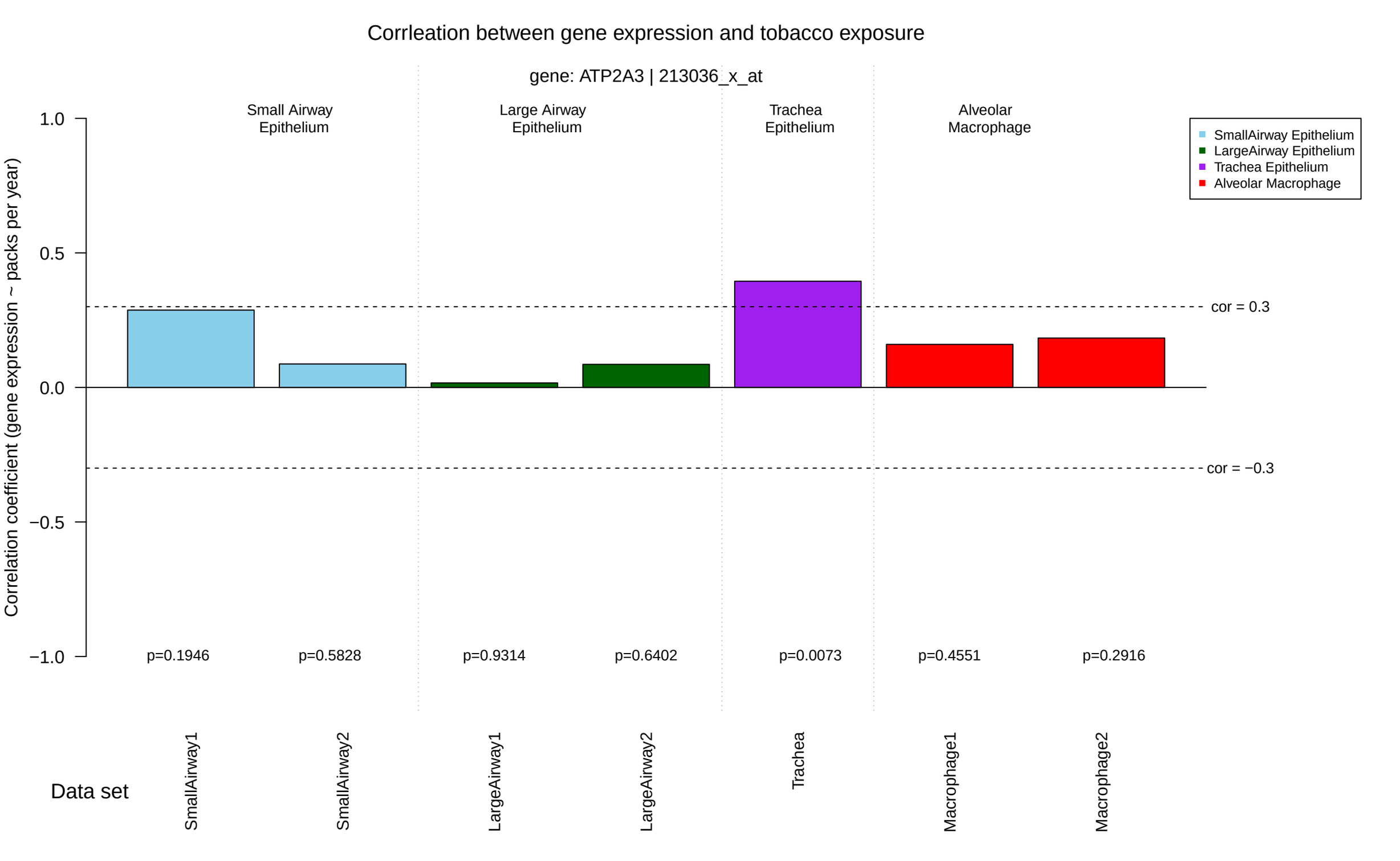

set: LargeAirway1 (p=0.4796)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.115)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0014)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5059)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0925)
ATP2A3 | 213036_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP2A3 | 213036_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1946
p=0.5828
p=0.9314
p=0.6402
p=0.0073
p=0.4551
p=0.2916
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2